This assignment will not be graded and is only for practice.

Level 1:

1) Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as $f(x, y) = x^2 + 2xy^2 + 2x^2y + y^2$. If $\nabla f(1, 1) = (a, b)$, then find the value of $a - b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

2) Consider the following function defined as

$$f_1(x, y) = \begin{cases} \frac{x}{y} & \text{if } y = 0 \\ 1 & \text{if } y = 0. \end{cases}$$

$$f_2(x, y) = xy + x^2y^2$$

Statement 1: f_1 has continuous gradient at the origin but f_2 does not have continuous gradient at the origin.
Statement 2: f_2 has continuous gradient at the origin but f_1 does not have continuous gradient at the origin.
Statement 3: $f_1(x, y)$ and $f_2(x, y)$ have continuous gradients at the origin.
Which of the above statements is correct?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

3) Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as $f(x, y) = e^{(x^2+xy+y^3)}$. ***1 point***

Which of the following option(s) is (are) true?

- [ ] $\nabla f = (e^{(x^2+xy+y^3)}, e^{(x^2+xy+y^3)})$
- [ ] $\nabla f = (2x + y, 3y^2 + x)$
- [ ] $\nabla f = (2xe^{(x^2+xy+y^3)}, 3y^2e^{(x^2+xy+y^3)})$

- [ ] $f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point (1,1).
- [ ] $f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point (1,-1).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point (1,1).
$f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point (1,-1).

Level 2:

With a particular frame of reference (in R^2), a corner of a hot rectangular iron plate with some finite length and breadth is placed at the origin $(0, 0)$. The temperature of the plate at a point (x, y) is given by the function:

$$T(x, y) = xy^2 + x^2y$$

Use this information to answer the following questions

4) From the point (1,3), in which direction the rate of change in temperature is the minimum? ***1 point***

- $\frac{1}{\sqrt{274}}$(-15, -7)
- $\frac{1}{\sqrt{82}}$(9, 1)
- $\frac{1}{\sqrt{82}}$(-9, -1)
- $\frac{1}{\sqrt{274}}$(15, 7)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{274}}$(-15, -7)

5) From the point (1,3), in which direction the rate of change in temperature is the maximum? ***1 point***

- $\frac{1}{\sqrt{274}}$(-15, -7)
- $\frac{1}{\sqrt{82}}$(9, 1)
- $\frac{1}{\sqrt{82}}$(-9, -1)
- $\frac{1}{\sqrt{274}}$(15, 7)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{274}}$(15, 7)

6) From a point (1,3), in which direction the rate of change in temperature of the iron plate is zero? ***1 point***

- [ ] $\frac{1}{\sqrt{274}}$(-7, 15)
- [ ] $\frac{1}{\sqrt{82}}$(1, -9)
- [ ] $\frac{1}{\sqrt{82}}$(-1, 9)
- [ ] $\frac{1}{\sqrt{274}}$(7, -15)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{274}}$(-7, 15)
$\frac{1}{\sqrt{274}}$(7, -15)

7) Consider a function $f : \mathrm{R}^2 \rightarrow \mathrm{R}$ defined as $f(x, y, z) = 2e^{2x} \sin yz$. At the point (0, $\frac{\pi}{2}$, 1), which are the directions that have the directional derivative to be 0 ? ***1 point***

- [ ] $\frac{1}{\sqrt{2}}$ (0,1,1)
- [ ] $\frac{1}{\sqrt{2}}$ (0,1,-1)
- [ ] $\frac{1}{\sqrt{3}}$ (1,1,1)
- [ ] $\frac{1}{2021\sqrt{2}}$ (0,2021,-2021)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{2}}$ (0,1,1)
$\frac{1}{\sqrt{2}}$ (0,1,-1)
$\frac{1}{2021\sqrt{2}}$ (0,2021,-2021)

8) Consider a function $f : \mathrm{R}^3 \rightarrow \mathrm{R}$ defined as $f(x, y, z) = \alpha x + \beta xy^2 + \gamma yz^2$. If the maximum value of the directional derivative of $f(x, y, z)$ at (1, 1, 1) is in the direction parallel to x- axis and has magnitude 5, then ***1 point***

$f(x, y, z) = 5x + x^2$

- ○ $f(x, y, z) = 2x + xy^2$
- ○ $f(x, y, z) = 5x$
- ○ $f(x, y, z) = xy^2 + 2yz^2$
- ○ $f(x, y, z) = 2x$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y, z) = 5x$

Your score is: 0/8